

16. Suppose a die is rolled n times. Show that the probability of a roll of i appearing k_i times, $i = 1, \dots, 6$, is

$$\phi(k_1, \dots, k_6) = \frac{n!}{k_1! \dots k_6!} \frac{1}{6^n}.$$

The expected value of the total of n rolls is $3.5n$. Suppose $\sum_i i k_i = \rho n$, $1 \leq \rho \leq 6$, and $k_i = n p_i$. Use Stirling's formula to show that subject to these constraints, ϕ is maximized by choosing the p_i as nonnegative numbers that solve the optimization problem

$$\text{maximize } -\sum_i p_i \log p_i, \quad \text{subject to } \sum_i p_i = 1, \text{ and } \sum_i i p_i = \rho.$$

We only do this

For $\rho = 4$ the maximizer is $p^* \approx (0.103, 0.123, 0.146, 0.174, 0.207, 0.247)$, so the most likely way of obtaining a total of $4n$ is when about 24.7% of the dice rolls are a 6. What do you guess would be the solution to the optimization problem if $\rho = 3.5$? If $\rho = 3$?

$$\phi(k_1, \dots, k_6) = \frac{n!}{k_1! \dots k_6!} \frac{1}{6^n}$$

① Use Strong Form of Stirling's Formula

$$\phi(k_1, \dots, k_6) \sim \frac{\sqrt{2\pi n} \left(\frac{n}{e}\right)^n}{\prod_{i=1}^6 \sqrt{2\pi k_i} \left(\frac{k_i}{e}\right)^{k_i}} \frac{1}{6^n} \quad \because p_i = \frac{k_i}{n}$$

$$= \frac{\sqrt{2\pi}}{(\sqrt{2\pi})^6} \cdot \frac{\sqrt{n} \left(\frac{n}{e}\right)^n \cdot e^{\overbrace{k_1+k_2+\dots+k_6}^{=n}}}{\sqrt{k_1 \dots k_6} k_1^{k_1} \dots k_6^{k_6}} \frac{1}{6^n} \quad \sum_i p_i = 1$$

$$= \frac{1}{(\sqrt{2\pi})^5} \frac{\sqrt{n} n^n}{\sqrt{k_1 \dots k_6} k_1^{k_1} \dots k_6^{k_6}} \frac{1}{6^n}$$

$$= \frac{1}{(\sqrt{2\pi})^5} \frac{n^{n+\frac{1}{2}}}{k_1^{k_1+\frac{1}{2}} k_2^{k_2+\frac{1}{2}} \dots k_6^{k_6+\frac{1}{2}}} \frac{1}{6^n}$$

$$= \frac{1}{(\sqrt{2\pi})^5 6^n} e^{(n+\frac{1}{2}) \log n - \left(\sum_i (k_i + \frac{1}{2}) \log k_i\right)}$$

$$= \frac{1}{(\sqrt{2\pi})^5 6^n} e^{(n+\frac{1}{2}) \log n - \left(\sum_i (n p_i + \frac{1}{2}) \log(n p_i)\right)}$$

To optimize ϕ , it suffices to optimize

$$-\sum (n p_i + \frac{1}{2}) \log(n p_i)$$

$$= -\sum n p_i \log(n p_i) - \frac{1}{2} \sum \log(n p_i)$$

$\underbrace{\log n + \log p_i}_{\text{same}}$

$$= -\sum n p_i \log n - \sum n p_i \log p_i - \frac{1}{2} \sum \log n$$

$\sum p_i = 1$

$$- \frac{1}{2} \sum \log p_i$$

So it suffices to optimize $-\sum n p_i \log p_i - \sum \frac{1}{2} \log p_i$

$$= -n \left(\sum p_i \log p_i + \frac{1}{2n} \sum \log p_i \right)$$

Since we are using asymptotic function, n is very large.

How to argue this term can be ignored??

② Use Weak Form of Stirling Formula

$$\log n! \sim n \log n.$$

$$\therefore \phi = \frac{n!}{k_1! \cdots k_b!} \cdot \frac{1}{6^n}$$

$$\log \phi = \log n! - \left(\sum_i \log k_i! \right) - \log 6^n$$

$$\sim n \log n - \left(\sum_i k_i \log k_i \right) - n \log 6$$

$$= n \log n - \sum_i n p_i \log (n p_i) - n \log 6$$

$$= n \log n - n \log 6 - \sum \underline{n p_i} \log n$$

$$- \sum n p_i \log p_i$$

$$= \underline{n \log n} - n \log 6 - \underline{n \log n}$$

$$- \sum n p_i \log p_i$$

$$= -n \log 6 - \underline{n \sum p_i \log p_i}$$

To optimize ϕ , it suffices to optimize $-\sum p_i \log p_i$

Constraints: $\sum k_i = n \Rightarrow \sum p_i = 1$

$\sum i k_i = \rho n \Rightarrow \sum i p_i = \rho$ [use $k_i = n p_i$]